

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Tenvir-EM

(Tenofovir Disoproxil Fumarate & Emtricitabine Tablets)

WARNING

LACTIC ACIDOSIS AND SEVERE HEPATOMEGALY WITH STEATOSIS, INCLUDING FATAL CASES, HAVE BEEN REPORTED WITH THE USE OF NUCLEOSIDE ANALOGS ALONE OR IN COMBINATION WITH OTHER ANTIRETROVIRALS (SEE WARNINGS AND PRECAUTIONS).

TENVIR-EM IS NOT INDICATED FOR THE TREATMENT OF CHRONIC HEPATITIS B VIRUS (HBV) INFECTION AND THE SAFETY AND EFFICACY OF TENVIR-EM HAVE NOT BEEN ESTABLISHED IN PATIENTS CO-INFECTED WITH HBV AND HIV. SEVERE ACUTE EXACERBATIONS OF HEPATITIS B HAVE BEEN REPORTED IN PATIENTS WHO HAVE DISCONTINUED EMTRICITABINE OR TENOFOVIR DF. HEPATIC FUNCTION SHOULD BE MONITORED CLOSELY WITH BOTH CLINICAL AND LABORATORY FOLLOW-UP FOR AT LEAST SEVERAL MONTHS IN PATIENTS WHO DISCONTINUE TENVIR-EM AND ARE CO-INFECTED WITH HIV AND HBV. IF APPROPRIATE, INITIATION OF ANTI-HEPATITIS B THERAPY MAY BE WARRANTED (SEE WARNINGS AND PRECAUTIONS).

Composition

Each film coated tablet contains

Tenofovir disoproxil fumarate 300 mg

equivalent to Tenofovir disoproxil 245 mg

Emtricitabine 200 mg

Description

Blue coloured capsule shaped biconvex film coated tablet plain on both sides.

Dosage Form

Tablets

PHARMACOLOGY

Pharmacodynamics

Emtricitabine is an NRTI that is converted intracellularly to its active metabolite, emtricitabine 5'-triphosphate [1] Tenofovir DF is an ester prodrug of the NRTI, tenofovir. [2] The prodrug is hydrolysed intracellularly to tenofovir, which is then converted to the active metabolite, tenofovir diphosphate. The active metabolites of the drugs compete with deoxycytidine 5'-triphosphate (emtricitabine 5'-triphosphate) or deoxyadenosine 5'-triphosphate (tenofovir diphosphate) for incorporation into HIV DNA, thereby terminating viral DNA chain growth and inhibiting the activity of the viral reverse transcriptase.

Pharmacokinetics in Adults

Absorption

The median bioavailability of emtricitabine in fasted patients following a single oral dose is 92%. The median bioavailability of tenofovir in fasted patients following a single oral dose is 25% [3].

Distribution

Protein binding of emtricitabine is less than 4%; and of tenofovir, is less than 0.7% (plasma) and 7.2% (serum) [3].

Metabolism and Elimination

Emtricitabine undergoes limited metabolism and is primarily excreted by the kidneys The median terminal elimination half-lives of emtricitabine and tenofovir were 15.5 and 17.6 hours when healthy volunteers received a single dose of co-formulated emtricitabine/tenofovir DF 200mg/300mg on two separate occasions [3].

INDICATIONS

Treatment of HIV-1 infection

TENVIR-EM is indicated in combination with other antiretroviral agents (such as non-nucleoside reverse transcriptase inhibitors or protease inhibitors) for the treatment of HIV-1 infection in adults.

The following points should be considered when initiating therapy with TENVIR-EM for the treatment of HIV-1 infection:

- It is not recommended that TENVIR-EM be used as a component of a triple nucleoside regimen.
- TENVIR-EM should not be co-administered with lamivudine + tenofovir DF + Efavirenz, emtricitabine, tenofovir DF or lamivudine-containing products

In treatment experienced patients, the use of TENVIR-EM should be guided by laboratory testing and treatment history.

Pre-exposure prophylaxis (PrEP)

Emtricitabine/Tenofovir Disoproxil is indicated in combination with safer sex practices for pre-exposure prophylaxis to reduce the risk of sexually acquired HIV-1 infection in adults at high risk. When prescribing TENVIR-EM for PrEP, healthcare providers must:

- prescribe TENVIR-EM as part of a comprehensive prevention strategy because emtricitabine/tenofovir disoproxil is not always effective in preventing the acquisition of HIV-1 infection.
- counsel all uninfected individuals to strictly adhere to the recommended TENVIR-EM dosing schedule because the effectiveness of emtricitabine/tenofovir disoproxil in reducing the risk of acquiring HIV-1 was strongly correlated with adherence as demonstrated by measurable drug levels in clinical trials.
- confirm a negative HIV-1 test immediately prior to initiating TENVIR-EM for a PrEP indication. If clinical symptoms consistent with acute viral infection are present and recent (<1 month) exposures are suspected, delay starting PrEP for at least one month and reconfirm HIV-1 status, including acute or primary HIV-1 infection; and - screen for HIV-1 infection at least once every 3 months while taking emtricitabine/tenofovir disoproxil for PrEP.

DOSAGE AND METHOD OF ADMINISTRATION

Treatment of HIV-1 infection

The dosage of TENVIR-EM is one tablet (containing 200 mg of emtricitabine and 300 mg of tenofovir DF) once daily taken orally with or without food.

Dose adjustment for renal impairment

Significantly increased drug exposures occurred when emtricitabine or tenofovir DF were administered to patients with moderate to severe renal impairment. Therefore, the dosing interval of TENVIR-EM should be adjusted in patients with baseline creatinine clearance of 30-49 mL/min using the recommendations in Table 1. The safety and effectiveness of these dosing interval adjustment recommendations have not been clinically evaluated; therefore, clinical response to treatment and renal function should be closely monitored in these patients.

No dose adjustment is necessary for patients with mild renal impairment (creatinine clearance of 50–80 mL/min). Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients with mild renal impairment.

Table 1: Dosage Adjustment for Patients with Altered Creatinine Clearance

	Creatinine Clearance (mL/min)^a		
	≥ 50	30-49	<30 (including patients requiring hemodialysis)
Recommended Dosing Interval	Every 24 hours	Every 48 hours	TENVIR-EM should not be administered

a Calculated using ideal (lean) body weight.

Pre-exposure prophylaxis (PrEP)

This indication is based on clinical trials in men who have sex with men (MSM) at high risk for HIV-1 infection and in heterosexual serodiscordant couples.

Posology

Prevention of HIV in adults: One tablet, once daily.

Adults with renal impairment

Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients with mild renal impairment.

TENVIR-EM should only be used in individuals with creatinine clearance (CrCl) < 80 mL/min if the potential benefits are considered to outweigh the potential risks. See Table 2.

Table 2: Dosing recommendations in adults with renal impairment

	Pre-exposure prophylaxis
Mild renal impairment (CrCl 50-80mL/min)	Limited data from clinical studies support once daily dosing in HIV-1 uninfected individuals with CrCl 60-80 mL/min. Use is not recommended in HIV-1 uninfected individuals with CrCl < 60 mL/min as it has not been studied in this population
Moderate renal impairment (CrCl 30-49 mL/min)	Not recommended for use in this population.

Severe renal impairment (CrCl < 30 mL/min) and haemodialysis patients.	Not recommended for use in this population.
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MODE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

TENVIR-EM is contraindicated in patients with previously demonstrated hypersensitivity to any of the components of the product.

Use for pre-exposure prophylaxis in individuals with unknown or positive HIV-1 status.

WARNINGS AND PRECAUTIONS

Overall HIV-1 infection prevention strategy

Emtricitabine/Tenofovir Disoproxil is not always effective in preventing the acquisition of HIV-1. The time to onset of protection after commencing emtricitabine/tenofovir disoproxil is unknown.

Emtricitabine/Tenofovir Disoproxil should only be used for pre-exposure prophylaxis as part of an overall HIV-1 infection prevention strategy including the use of other HIV-1 prevention measures (e.g., consistent and correct condom use, knowledge of HIV-1 status, regular testing for other sexually transmitted infections).

Risk of resistance with undetected HIV-1 infection:

Emtricitabine/Tenofovir Disoproxil should only be used to reduce the risk of acquiring HIV-1 in individuals confirmed to be HIV negative. Individuals should be re-confirmed to be HIV-negative at frequent intervals (e.g., at least every 3 months) using a combined antigen/antibody test while taking emtricitabine/tenofovir disoproxil for pre-exposure prophylaxis.

Emtricitabine/Tenofovir Disoproxil alone does not constitute a complete regimen for the treatment of HIV-1 and HIV-1 resistance mutations have emerged in individuals with undetected HIV-1 infection who are only taking emtricitabine/tenofovir disoproxil.

If clinical symptoms consistent with acute viral infection are present and recent (< 1 month) exposures to HIV-1 are suspected, use of emtricitabine/tenofovir disoproxil should be delayed for

at least one month and HIV-1 status reconfirmed before starting Emtricitabine/Tenofovir Disoproxil for pre-exposure prophylaxis.

Importance of adherence:

The effectiveness of emtricitabine/tenofovir disoproxil in reducing the risk of acquiring HIV-1 is strongly correlated with adherence as demonstrated by measurable drug levels in blood. HIV-1 uninfected individuals should be counselled at frequent intervals to strictly adhere to the recommended Emtricitabine/Tenofovir Disoproxil daily dosing schedule.

Lactic Acidosis/Severe Hepatomegaly with Steatosis

Lactic acidosis and severe hepatomegaly with steatosis, including fatal cases, have been reported with the use of nucleoside analogs alone or in combination with other antiretrovirals. Particular caution should be exercised when administering nucleoside analogs to any patient with known risk factors for liver disease; however, cases have also been reported in patients with no known risk factors. Treatment with TENVIR-EM should be suspended in any patient who develops clinical or laboratory findings suggestive of lactic acidosis or pronounced hepatotoxicity (which may include hepatomegaly and steatosis even in the absence of marked transaminase elevations).

Patients with HIV, HBV and HCV Co-infection

The efficacy and safety of tenofovir disoproxil fumarate-emtricitabine has not been thoroughly studied in the treatment of HIV-HBV co-infection, several studies have indicated that discontinuation of tenofovir disoproxil fumarate-emtricitabine in HBV infected patients may result in severe and acute exacerbation or flare up of the hepatitis. [5] Liver function tests should be monitored for at least several months in HIV/HBV co-infected patients who are suspended from treatment with tenofovir disoproxil fumarate emtricitabine. Because of the risk of HIV resistance with two- drug therapy, if co-infected patients require treatment, other agents such as adefovir or pegylated interferon should be used.

The safety and efficacy of Emtricitabine/Tenofovir Disoproxil for pre-exposure prophylaxis in patients with HBV or Hepatitis C Virus (HCV) infection has not been established.

New Onset or Worsening of Renal Impairment

Emtricitabine and tenofovir are principally eliminated by the kidney. Renal impairment, including cases of acute renal failure and Fanconi syndrome (renal tubular injury with severe hypophosphatemia), has been reported with the use of tenofovir DF.

It is recommended that creatinine clearance be calculated in all patients prior to initiating therapy and as clinically appropriate during therapy with tenofovir disoproxil fumarate and emtricitabine. Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients at risk for renal impairment.

Emtricitabine and tenofovir disoproxil fumarate should be avoided with concurrent or recent use of a nephrotoxic agent.

For the treatment of HIV-1 infection, dosing interval adjustment of TENVIR-EM and close monitoring of renal function are recommended in all patients with creatinine clearance 30-49 mL/min. No safety or efficacy data are available in patients with renal impairment who received tenofovir DF and emtricitabine using these dosing guidelines, so the potential benefit of TENVIR-EM therapy should be assessed against the potential risk of renal toxicity. TENVIR-EM should not be administered to patients with creatinine clearance <30 mL/min or patients requiring hemodialysis.

Renal management in pre-exposure prophylaxis:

Emtricitabine/Tenofovir Disoproxil has not been studied in HIV-1 uninfected individuals with creatinine clearance < 60 mL/min and is therefore not recommended for use in this population. If serum phosphate is < 1.5 mg/dL (0.48 mmol/L) or creatinine clearance is decreased to < 60 mL/min in any individual receiving emtricitabine/tenofovir disoproxil for pre-exposure prophylaxis, renal function should be re-evaluated within one week, including measurements of blood glucose, blood potassium and urine glucose concentrations. Consideration should be given to interrupting use of emtricitabine/tenofovir disoproxil in individuals with creatinine clearance decreased to < 60 mL/min or decreases in serum phosphate to < 1.0 mg/dL (0.32 mmol/L). Interrupting use of emtricitabine/tenofovir disoproxil should also be considered in case of progressive decline of renal function when no other cause has been identified.

Decreases in Bone Mineral Density

Subclinical renal phosphate wasting could possibly contribute to a decrease in bone mineral density that was reported in a study conducted among treatment-naïve patients who were treated with tenofovir disoproxil fumarate in reference to a published trial report [5]. A decrease in bone mineral density at the lumbar spine and hip, especially between weeks 24 and 48, was evident but was non-progressive through 288 weeks. Thus, monitoring of bone mineral density should be considered in patients with history of, or risk factors for, pathologic fractures. To the knowledge

of the authors, no study has examined the benefit of calcium and vitamin D supplementation or bisphosphonates in this setting [5].

Fat Redistribution

Redistribution/accumulation of body fat including central obesity, dorsocervical fat enlargement (buffalo hump), peripheral wasting, facial wasting, breast enlargement and “cushingoid appearance” have been observed in patients receiving antiretroviral therapy.

The mechanism and long-term consequences of these events are currently unknown. A causal relationship has not been established.

Immune Reconstitution Syndrome

Immune reconstitution syndrome has been reported in patients treated with combination antiretroviral therapy, including emtricitabine and tenofovir. During the initial phase of combination antiretroviral treatment, patients whose immune system responds may develop an inflammatory response to indolent or residual opportunistic infections (such as *Mycobacterium avium* infection, cytomegalovirus, *Pneumocystis jirovecii* pneumonia (PCP), or tuberculosis), which may necessitate further evaluation and treatment.

Early Virologic Failure

Caution when using triple nucleoside reverse transcriptase inhibitor (NRTI) therapy. Clinical studies have demonstrated that such treatments are generally less effective than triple drug regimens containing 2 NRTIs in combination with either a non– nucleoside reverse transcriptase inhibitor (NNRTI) or a protease inhibitor; early virologic failure and high rates of resistance substitutions have been reported. Patients receiving triple NRTI therapy should be carefully monitored and considered for treatment modification.[4].

This medicinal product contains lactose monohydrate. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose galactose malabsorption should not take this medicine.

DRUG INTERACTIONS

Any medications that reduce renal function may increase concentration of emtricitabine and tenofovir disoproxil fumarate. No drug interactions with clinical consequences are known for emtricitabine. Co-administration of tenofovir disoproxil fumarate with certain protease inhibitors, including atazanavir, lopinavir/ritonavir, and darunavir (TMC-114)/ritonavir results in increased plasma levels of tenofovir disoproxil fumarate and, in the case of atazanavir and lopinavir/

ritonavir, reduced protease inhibitor troughs. While no dose adjustment is recommended when tenofovir disoproxil fumarate is used with these drugs, in cases where baseline protease resistance mutations have modestly raised the concentration of the drug required to inhibit the virus, this effect of tenofovir disoproxil fumarate could be significant. Atazanavir should always be ritonavir boosted (atazanavir 300 mg should be boosted with ritonavir 100 mg) when used with tenofovir disoproxil fumarate[5].

Tenofovir disoproxil fumarate should be used very cautiously with didanosine due to increased rates of adverse reactions including peripheral neuropathy and pancreatitis. This is likely a result of a drug interaction resulting in a 40%–50% increase in plasma didanosine levels and/or intracellular drug interactions. In addition, use of tenofovir disoproxil fumarate with didanosine has also been associated with paradoxical CD4 declines or less than robust CD4 increases in some, but not all studies. These results are reminiscent of those seen in studies combining didanosine with hydroxyurea. If co-administration of didanosine and tenofovir disoproxil fumarate is necessary, didanosine dose should be adjusted and patients should be closely monitored for didanosine-related adverse reactions. The recommended didanosine dose for patients weighing >60 kg in this setting is 250 mg; however, there is no adequate information for patients weighing below 60 kg [5].

Patients with Renal Impairment

Because both tenofovir disoproxil fumarate and emtricitabine are excreted via the kidneys, modification of tenofovir disoproxil fumarate dosage and special caution is necessary when treating patients with renal impairments.

PREGNANCY

Pregnancy Category B:

No adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, TENVIR-EM should be used during pregnancy only if clearly needed.

LACTATION

Tenofovir disoproxil fumarate-Emtricitabine is excreted in breast milk and should not be used while nursing. The Centers for Disease Control and Prevention recommend that HIV-infected women do not breast feed their infants to avoid risking postnatal transmission of HIV.

UNDESIRABLE EFFECTS

In reference to published trial report [6] antiretroviral-naïve patients received emtricitabine, tenofovir disoproxil fumarate and efavirenz for 144 weeks, the most frequently reported adverse reactions considered possibly or probably related to emtricitabine and/or tenofovir disoproxil fumarate were nausea (12%) and diarrhoea (7%).

Grade 2 through 4 laboratory abnormalities occurred in 67% of patients in the Tenofovir disoproxil fumarate/emtricitabine arm compared with 68% in the ZDV/3TC arm. Only 3% and 4%, respectively, of these laboratory abnormalities occurred between 96 and 144 weeks.

Grade 3 and 4 laboratory abnormalities occurring in 5% or greater of patients in the tenofovir disoproxil fumarate/emtricitabine or ZDV/3TC arm included elevated creatine kinase (9% vs. 7%), triglycerides (5% vs. 3%), neutropenia (3% vs. 5%), and amylase (8% vs. 4%) [6].

OVERDOSAGE

If overdose occurs, the patient must be monitored for evidence of toxicity, and standard supportive treatment applied as necessary.

SHELF LIFE

24 months

STORAGE CONDITION

Store below 30°C.

PRESENTATION

Plastic container of 30 tablets.

PRODUCT REGISTRATION HOLDER

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